

Retail Market Basket Intelligence

Phase 1: Exploratory Data Analysis & Data Discovery Summary

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Tools Deployed:

- MySQL Workbench
- Microsoft Excel
- Python (Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn)

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1. EXECUTIVE SUMMARY

This report presents the Phase 1 deliverables for the Retail Market Basket Intelligence pipeline, focusing on understanding the underlying dataset and conducting comprehensive Exploratory Data Analysis (EDA).

The analysis was performed on a retail transaction dataset containing 522,064 records and 7 columns, representing individual product purchases within customer transactions. To ensure rigorous data validation, the dataset was analyzed using three distinct analytical technologies:

- **MySQL Workbench:** For SQL-based dataset exploration, schema validation, and aggregate profiling.
- **Microsoft Excel:** For rapid pivot-table-based exploration and visual distribution checks.
- **Python (Jupyter Notebook):** For advanced statistical analysis, outlier detection, and programmatic visualization.

Key insights discovered during Phase 1 include:

- The dataset contains 21,663 unique transactions, 4,186 unique products, and 4,298 unique customers.
- The average purchase quantity per record is 10.09 units, while the median quantity is 3 units, indicating a highly right-skewed distribution.
- The dataset contains critical data quality gaps: 134,041 missing CustomerID values (25.68%) and 1,455 missing Itemname values (0.28%).
- 5,286 exact duplicate rows were detected.
- Extremely large outliers were identified, including a maximum quantity of 80,995 units and a minimum quantity of -9,600, indicating possible returns or bulk wholesale transactions.

The exploratory analysis revealed important patterns related to product demand, customer behavior, and transaction characteristics, establishing the blueprint for the next phase of the project: Data Cleaning and Feature Transformation.

2. PROJECT OVERVIEW

2.1 Business Problem

Retail businesses collect large volumes of transactional data through daily sales. Extracting actionable insights from this data can reveal highly valuable information about customer purchasing behavior and product relationships. This project aims to leverage Market Basket Analysis (MBA) to mathematically uncover product associations within raw transaction data.

2.2 Purchase Pattern Analytics Concept

Purchase Pattern Analytics analyzes transaction data to formally understand:

- Which products are frequently purchased together.
- How customers organically build their shopping baskets.
- Which products act as anchors to drive higher overall sales.

2.3 Importance of Market Basket Analysis

Deploying Market Basket Analysis helps retail enterprises:

- Identify product combinations that are frequently purchased together.
- Design profitable product bundles and targeted promotions.
- Optimize physical and digital store layouts.
- Exponentially increase cross-selling opportunities.

3. DATASET OVERVIEW

The dataset analyzed contains historical retail transaction records.

3.1 Dataset Size

- **Total Records:** 522,064
- **Total Columns:** 7

3.2 Dataset Schema

Column	Description
BillNo	Unique transaction identifier
Itemname	Name of the product purchased
Quantity	Number of units purchased per line item
Present_Date	Date and time of the purchase
Price	Unit price of the product
CustomerID	Unique identifier of the purchasing customer
Country	Geographic country where the purchase occurred

3.3 Dataset Characteristics

- 21,663 unique transactions.
- 4,186 unique products.
- 4,298 unique customers.
- 30 distinct countries represented in the dataset.
- *Conclusion:* This dataset structure is highly suitable for transaction-level retail analytics and association rule mining.

4. ANALYTICAL METHODOLOGY

Phase 1 methodology focused on extensive data discovery and initial quality assessment across three industry-standard toolsets:

- **SQL (MySQL Workbench):** Structured dataset exploration and validation, identification of hidden missing values, duplicates, column audits, and aggregate statistics.
- **Microsoft Excel:** Filtering, pivot-table analyses, and quick visualization checks for distribution and concentration patterns.
- **Python (Jupyter Notebook):** Full data profiling using Pandas, statistical summaries, distribution plots (histograms, KDE), boxplots, grouping/aggregation, correlation analysis, and outlier/anomaly detection (Z-score).

5. EXPLORATORY DATA ANALYSIS (TOOL-BY-TOOL BREAKDOWN)

5.1 EDA Using MySQL

SQL queries were deployed to perform structured, database-level exploration.

5.1.1 Dataset Inspection

- Total rows: 522,064 | Total columns: 7
- Distinct transactions: 21,663
- Distinct products: 4,186
- Distinct customers: 4,298

5.1.2 Column Audit Findings (Data Types)

- BillNo → Text
- Itemname → Text
- Quantity → Integer
- Present_Date → Text/Date
- Price → Float
- CustomerID → Text
- Country → Text

5.1.3 Missing Values Detection

Initial SQL analysis revealed missing values were deceptively stored as blank strings ("") rather than native NULL values.

- CustomerID: 134,041 missing records
- Itemname: 1,455 missing records

5.1.4 Duplicate Records

SQL duplicate detection identified **5,286 duplicate rows**. If untreated, these duplicates will artificially inflate product counts and transaction statistics.

5.1.5 Quantity Validation

- Mean quantity: 10.09
- Median quantity: 3
- Mode quantity: 1
- Minimum quantity: -9,600 (*Negative quantities indicate product returns*)
- Maximum quantity: 80,995
- Standard deviation: 161.11

5.1.6 Price Validation

- Mean price: 3.83
- Minimum price: -11,062.06 (*Negative prices likely represent refunds or ledger corrections*)
- Maximum price: 13,541.33

5.1.7 Product Frequency Analysis (Top products by total volume)

- PAPER CRAFT, LITTLE BIRDIE → 80,995 units
- MEDIUM CERAMIC TOP STORAGE JAR → 77,553 units
- WORLD WAR 2 GLIDERS ASSTD DESIGNS → 53,799 units
- JUMBO BAG RED RETROSPOT → 48,002 units
- WHITE HANGING HEART T-LIGHT HOLDER → 36,527 units

5.1.8 Customer Analysis (Most active customers by transaction count)

- Customer 12748 → 210 transactions
- Customer 17841 → 124 transactions
- Customer 13089 → 97 transactions
- Customer 14606 → 93 transactions
- Customer 15311 → 91 transactions

5.1.9 Transaction Analysis (Basket sizes)

- Average products per transaction: 24.03
- Median basket size: 13 products
- Maximum basket size: 1,114 products

5.2 EDA Using Microsoft Excel

Excel was utilized for rapid visual and pivot-based aggregation.

- **Filtering and Sorting:** Confirmed missing CustomerID values in 25.68% of rows and missing Itemname values in 0.28% of rows.
- **Pivot Table Results:** Confirmed the exact same top-selling products (Paper Craft, Ceramic Jar, Jumbo Bag) found in the SQL and Python pipelines, validating data consistency.
- **Customer Activity Patterns:** Pivot tables proved that a very small group of customers accounts for a disproportionately large portion of transactions (e.g., Customer 12748 with 210 purchases).
- **Transaction Size Patterns:** Pivot analysis revealed the vast majority of standard transactions contain between 4 and 27 items, though severe outliers exceed 1,000 items.
- **Visualization Insights:** Excel charts visually confirmed the highly skewed quantity distribution, the dominance of specific flagship products, and a massive geographic concentration of transactions originating in the United Kingdom.

5.3 EDA Using Python

Python provided deep statistical modeling and programmatic visualization.

- **Dataset Structure:** Shape confirmed at (522,064 rows, 7 columns).
- **Unique Value Counts:** 21,663 transactions, 4,186 products, 4,298 customers, 30 countries.
- **Missing Values:** Python cleanly mapped hidden blanks to NaNs, confirming 134,041 missing CustomerIDs (25.68%) and 1,455 missing Itemnames (0.28%).
- **Duplicate Records:** 5,286 rows detected via `df.duplicated()`.
- **Statistical Distribution Analysis:** Histogram and KDE plots proved the Quantity distribution is heavily right-skewed, with the standard consumer majority purchasing strictly between 1-10 units.
- **Outlier & Anomaly Detection:** Boxplot and Z-score methods detected exactly **539 extreme quantity outliers** that skew aggregate metrics.
- **Product & Customer Behavior:** Matched SQL findings precisely (Paper Craft maxing at 80,995 units; Customer 12748 maxing at 210 transactions).
- **Correlation Analysis:** The Pearson correlation coefficient between Quantity and Price was calculated at **-0.0028**. This mathematically proves there is almost no relationship between the unit price and the quantity purchased.
- **Grouped Statistics by Country:**
 - Netherlands → Average quantity: 85 units
 - Japan → Average quantity: 81 units
 - United Kingdom → Average quantity: 9.28 units

6. KEY INSIGHTS FROM EDA

- The dataset contains 522,064 retail transaction records.
- There are 21,663 unique transactions.
- The dataset includes 4,186 unique products.
- There are 4,298 customers in the dataset.
- 25.68% of CustomerID values are missing.
- 1,455 product names are missing.
- 5,286 duplicate rows were identified.
- Average quantity purchased per record is 10.09 units.
- Median quantity purchased is 3 units.
- Maximum quantity recorded is 80,995 units.
- Some transactions contain over 1,000 products.
- Product demand distribution is highly skewed.
- The United Kingdom accounts for the majority of transactions.
- Price and quantity show almost no correlation (-0.0028).
- Certain flagship products completely dominate total sales volume.

7. DATA QUALITY OBSERVATIONS

The EDA uncovered critical roadblocks that must be resolved before algorithmic modeling:

1. **Missing Customer IDs:** 134,041 records missing (25.68%).
 - *Implication:* Limits deeper customer behavior analysis.
2. **Missing Product Names:** 1,455 missing values.
 - *Implication:* Prevents mathematical product association analysis.
3. **Duplicate Records:** 5,286 duplicates detected.
 - *Implication:* Duplicates artificially inflate sales statistics and confidence metrics.
4. **Negative Quantities:** Minimum quantity = -9,600.
 - *Implication:* Indicates product returns or order cancellations, not purchases.
5. **Extreme Values:** Maximum quantity = 80,995.
 - *Implication:* Represents bulk wholesale purchases or data entry errors that skew consumer analytics.

(Note: These issues dictate the strict data cleansing pipeline required for Phase 2).

8. DOMAIN ANALYSIS

Retail data analytics focuses on extracting commercial insights from transactional data.

8.1 Customer Purchase Behavior

Understanding purchase patterns helps identify loyal customers, standard buying habits, and organic product preferences.

8.2 Market Basket Analysis

Market Basket Analysis identifies products that customers frequently purchase together (e.g., discovering mathematically that customers who buy bread possess a high probability of buying butter).

8.3 Product Co-Occurrence

Identifying product combinations helps retailers increase cross-selling efficiency, design high-conversion promotions, and optimize physical store layouts.

8.4 Business Value

Advanced retail analytics directly empowers businesses to increase overall revenue, improve inventory management, and enhance the digital/physical customer experience.

9. PREPARATION FOR PHASE 2: DATA ENGINEERING

The Phase 1 analysis has successfully mapped the dataset and isolated all anomalies, preparing the data for the next phase. Next steps include:

9.1 Data Cleaning Pipeline

- Remove all duplicate rows.
- Handle/impute missing values.
- Filter out inconsistent entries (negative quantities and negative prices).

9.2 Feature Transformation

- Convert raw transactional data into the sparse binary basket format strictly required for algorithmic Market Basket Analysis.

9.3 Algorithmic Deployment

- Deploy the Apriori Algorithm against the cleaned matrix to generate frequent itemsets, extract association rules, and formulate business-ready product recommendations.